

MADHESH P D

Full Stack Developer | AWS Cloud & Backend Engineer

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PROFESSIONAL SUMMARY

Full Stack Developer with hands-on production experience building and deploying scalable web applications on AWS. Specialises in Node.js backend services, React.js single-page applications (SPAs), and serverless architectures using AWS Lambda, API Gateway, Amazon Cognito, Amazon SQS, and Amazon DynamoDB. Comfortable owning the full delivery lifecycle — from database schema design to CDN deployment — across multiple concurrent products in a fast-paced startup environment. Experienced with TypeScript, CI/CD pipelines, and role-based access control (RBAC).

PROFESSIONAL EXPERIENCE

Full Stack Developer | **Pitfall Pvt Ltd** | Tiruchirappalli | March 2025 – Present

- Engineered Node.js and TypeScript backend services for 4 production applications (Business Management System, WhatsApp Messenger, Gold Pledge & Repledge, Restaurant Table Booking) on AWS EC2 with environment-based config and auto-scaling, maintaining consistent uptime across all environments.
- Architected serverless REST APIs with AWS Lambda and API Gateway, eliminating fixed server costs and enabling on-demand scaling for business-critical workflows.
- Deployed React.js single-page applications (SPAs) across all 4 products via AWS Amplify CI/CD pipeline, S3 static hosting, CloudFront CDN, and Route 53, reducing global page load times through edge caching.
- Designed multi-database architectures combining PostgreSQL (billing, loans, orders) with Amazon DynamoDB (high-throughput, low-latency operations), selecting storage per workload type rather than defaulting to one solution.
- Implemented Amazon Cognito for user authentication, JSON Web Token (JWT)-based authorisation, and role-based access control (RBAC) across all 4 platforms.
- Integrated Amazon SQS message queuing to decouple bulk WhatsApp notification delivery, enabling asynchronous processing and preventing bottlenecks under high-load scenarios.

Web Developer Intern | **Code Bind** | Tiruchirappalli | June 2023 – July 2023

- Built 3 full-stack applications (Car Rental System, Task Management, Tour Booking) end-to-end: user interface (UI) design, RESTful API development, and database integration using React.js, Node.js, and MySQL/PostgreSQL.
- Designed JSON Web Token (JWT)-based authentication and admin dashboard features covering real-time vehicle availability, booking management, task assignment, and itinerary browsing.
- Participated in Agile code reviews and maintained Git version control practices across all repositories.

PROJECTS

Task Management System | [View Project](#) | August 2024 – October 2024

- Full-stack application with React.js frontend (Netlify) and Node.js backend (Render) enabling real-time task tracking across distributed teams.
- Built a dual-role permission system — managers create, assign, and monitor tasks via a dashboard; team members view, update, and complete their assigned work.
- Developed CRUD-compliant RESTful APIs with Express.js and a relational database, covering user profiles, task records, assignments, priorities, deadlines, and status workflows.
- Implemented JSON Web Token (JWT) authentication with session management supporting both roles, enforcing data isolation between manager and team member accounts.

Agri-Bot — AI Agricultural Assistant | [View Project](#) | November 2024 – February 2025

- Built an AI-powered assistant integrating with NPK sensor hardware to capture real-time soil pH, nitrogen, phosphorus, and potassium readings for data-driven crop recommendations.

- Developed a machine learning (ML) crop recommendation engine that analyses sensor readings to suggest optimal crops for detected soil profiles, reducing crop failure risk for small-scale farmers.
- Delivered a unified Python full-stack platform connecting the NPK data pipeline, recommendation logic, and a government agricultural scheme discovery module into a single interface.

Car Rental System | [View Project](#) | 2023

- Full-stack car rental platform featuring real-time vehicle availability, secure online payments, and a comprehensive admin dashboard. Engineered backend APIs for booking management and integrated a cloud-hosted database, resulting in a 20% increase in online bookings and improved operational efficiency.

TECHNICAL SKILLS

Frontend: React.js, JavaScript (ES6+), TypeScript, HTML5, CSS3

Backend: Node.js, Express.js, TypeScript, RESTful API Design, Serverless Architecture, Microservices

AWS: EC2, Lambda, API Gateway, Amplify, S3, CloudFront, Route 53, Cognito, SQS, DynamoDB, IAM

Databases: PostgreSQL, MySQL, MongoDB, Amazon DynamoDB

DevOps & Tools: Git, GitHub, CI/CD (Amplify)

Other: JWT Authentication, Role-Based Access Control (RBAC), REST API Design

EDUCATION

B.E. – Computer Science and Engineering (First Class) | 2021 – 2025

Hindusthan Institute of Technology, Coimbatore | CGPA 7.21

Higher Secondary Certificate (HSC) | 2020 – 2021

Lakshmi Matric Higher Secondary School | 87%

Secondary School Leaving Certificate (SSLC) | 2018 – 2019

Lakshmi Matric Higher Secondary School | 83%